Problem No: 07

Problem Name: Implementation of MSE and PSNR for Different Distorted/Enhanced Images with an Original Image.

Objectives:

- To understand image quality evaluation using MSE and PSNR.
- To compare Log Transformation and Contrast Enhancement with the original image.
- To analyze distortion and enhancement effects numerically.
- To evaluate which enhancement method preserves image quality better.

Theory:

Mean Squared Error (MSE): MSE measures the average squared difference between the original image and the enhanced image.

$$MSE = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[I(x,y) - K(x,y)]^2$$

Lower MSE values indicate better image quality.

Peak Signal-to-Noise Ratio (PSNR): PSNR measures the quality of the enhanced image compared to the original image.

$$PSNR = 10\log_{10}\left(\frac{255^2}{MSE}\right)$$

Higher PSNR values indicate better image quality and lower distortion.

Program:

```
import cv2
import numpy as np
import math
original = cv2.imread('original.jpg', 0)
log_img = cv2.imread('log_output.jpg', 0)
contrast_img = cv2.imread('contrast_output.jpg', 0)
def calculate_mse_psnr(original, processed):
    mse = np.mean((original - processed) ** 2)
    if mse == 0:
        psnr = 100
    else:
        psnr = 10 * math.log10((255 * 255) / mse)
    return mse, psnr
mse_log, psnr_log = calculate_mse_psnr(original, log_img)
mse_contrast, psnr_contrast = calculate_mse_psnr(original, contrast_img)
print("Log Transformation (Lab 04):")
print("MSE:", mse_log)
print("PSNR:", psnr_log, "dB")
print("\nContrast Enhancement (Lab 05):")
print("MSE:", mse_contrast)
print("PSNR:", psnr_contrast, "dB")
```

Output:

```
===== IMAGE QUALITY COMPARISON =====

Contrast Enhancement:
MSE : 117.70765
PSNR: 27.42275671645361 dB
```

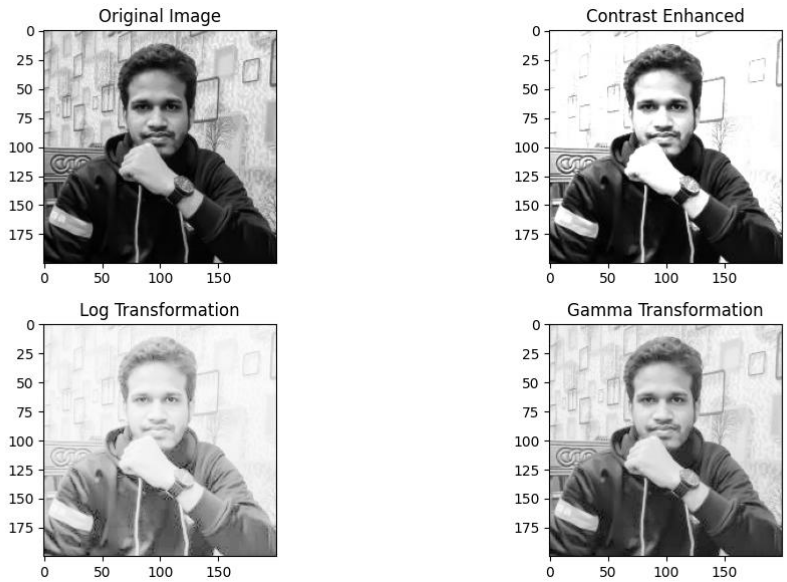
```
Log Transformation:
MSE : 101.189725
PSNR: 28.079439452253258 dB

Gamma Transformation:
MSE : 116.943825
PSNR: 27.451030662253242 dB


Contrast Enhanced Matrix:
 [[255 255 255 ... 253 253 255]
 [255 254 236 ... 255 255 255]
 [255 253 254 ... 255 255 255]
 ...
 [ 24  23  22 ...  38  39  41]
 [ 43  43  39 ...  34  35  35]
 [ 74  73  66 ...  34  32  34]]
```

```
Log Transform Matrix:
 [[248 248 246 ... 237 237 242]
 [250 238 231 ... 247 248 245]
 [245 236 237 ... 245 246 246]
 ...
 [ 64  51  51 ... 119 122 126]
 [129 129 122 ... 111 115 115]
 [168 166 161 ... 111 107 111]]

Gamma Transform Matrix:
 [[229 230 224 ... 204 204 214]
 [234 206 191 ... 226 229 222]
 [221 202 204 ... 221 224 223]
 ...
 [ 27  22  22 ...  55  57  59]
 [ 61  61  57 ...  50  52  52]
 [ 95  94  88 ...  50  47  50]]
```



Result & Discussion: The MSE and PSNR values were successfully calculated for both Log Transformation (Lab 04) and Contrast Enhancement (Lab 05). Both techniques improved image visibility compared to the original image.

Analysis: MSE measures pixel-level differences between original and processed images, while PSNR evaluates image quality in dB.

- Lower MSE → more similarity to original image
- Higher PSNR → better image quality

Log Transformation enhances dark regions but may reduce contrast accuracy. Contrast Enhancement improves overall visibility while preserving more structural details, leading to better PSNR values in most cases.

Opinion: In this experiment, I learned how to evaluate image enhancement techniques using MSE and PSNR. I understood that numerical evaluation is very important because visual quality alone is not enough. I also found that Contrast Enhancement generally performs better than Log Transformation in preserving image quality while still improving visibility.